

Winding-Free Exact Diagonalization of Quantum Spin Ice

Z.-B. Zhou¹

¹*Department of Physics, [institution], [city], [country]*
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Extracting quantum-spin-ice dynamics from thermodynamics requires finite clusters precisely where periodic boundaries generate a nonbulk process. On the 32-site pyrochlore torus, a four-spin loop returns to the ice manifold at order $J_{\pm}^2/J_{zz}$, parametrically before the physical six-spin ring at order $|J_{\pm}|^3/J_{zz}^2$. We assign every virtual sequence a gauge-invariant transported dipole and project its zero translation character before diagonalization. Through third order, this removes all 24 wrapping four-loops and 96 wrapping hexagons while preserving all 32 contractible hexagons and their amplitudes. An equivalent polarization mask makes the construction directly executable on the 2,970-state FCC-32 ice manifold. Operator counts, spectral collapse, heat capacity, and dynamical correlations establish that the projection removes the lower-order channel without suppressing physical ring dynamics. For $\text{Ce}_2\text{Hf}_2\text{O}_7$, the published linked-cluster parameters place the clean ring peak at 7.1 mK rather than the observed 25 mK. Enforcing both scales yields a new high-temperature-moment-preserving parameter target for a joint refit.

The two heat-capacity maxima of $\text{Ce}_2\text{Hf}_2\text{O}_7$, near 65 and 25 mK, sharpen a general numerical problem in quantum spin ice (QSI) [1]. Numerical linked-cluster expansion (NLC) constrains the exchange parameters from the higher-temperature line shape, whereas the lower feature lies at the scale where tunneling within the ice manifold should become coherent. In QSI, the ice rule is an emergent Gauss law and hexagonal ring exchange generates the low-energy $U(1)$ gauge dynamics [2–4]. Relating the two temperature scales therefore requires a controlled low-energy calculation. This is particularly important for the dipolar–octupolar pyrochlores, where candidate $U(1)_{\pi}$ regimes are not accessible to sign-free quantum Monte Carlo and exact diagonalization (ED) remains one of the few unbiased dynamical tools [5–9].

The standard 32-site periodic cluster is not innocuous in this regime. Its torus contains short alternating paths that do not exist in the infinite pyrochlore lattice. Most seriously, a four-spin path closes through a periodic identification and mixes ice configurations one perturbative order before the physical hexagonal ring. The error consequently grows relative to the desired signal as the ice expansion becomes more accurate. Neither increasing spectral resolution nor averaging observables over boundary twists identifies the offending virtual process. What is needed is an operator-level criterion that distinguishes bulk-contractible from boundary-wrapping transport.

Here we derive such a criterion and turn it into a finite-cluster protocol. Every virtual exchange sequence carries a translation character measured by its transported spin dipole. Its zero-character component is precisely the part that can be embedded without winding. We first expose the finite-size failure, then prove that this projection deletes the spurious four-site channel while retaining the physical six-site ring with unchanged coefficient. Independent matrix-element, thermodynamic, and spectroscopic tests establish that the operation cleans the topology rather than the physics. We finally combine the resulting

low-energy scale with the NLC constraint on $\text{Ce}_2\text{Hf}_2\text{O}_7$.

We use gauge-adapted axes and begin with

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+), \quad J_{zz} > 0. \quad (1)$$

Up to a constant, the Ising term is $H_0 = (J_{zz}/2) \sum_t Q_t^2$, where $Q_t = \sum_{i \in t} S_i^z$. The ice manifold $\mathcal{I}$ has $Q_t = 0$. A single term in V creates two charged tetrahedra, so $PVP = 0$. On the infinite lattice the shortest alternating cycle is a hexagon. Three exchanges annihilate the intermediate charges and produce the leading off-diagonal matrix element

$$g_{\text{hex}} = 12|J_{\pm}|^3/J_{zz}^2. \quad (2)$$

The $2 \times 2 \times 2$ primitive FCC torus instead contains 24 four-site cycles, all noncontractible. Two exchanges along either perfect matching return to $\mathcal{I}$ with amplitude

$$g_4 = 4J_{\pm}^2/J_{zz}, \quad \frac{g_4}{g_{\text{hex}}} = \frac{J_{zz}}{3|J_{\pm}|}. \quad (3)$$

Thus the torus artifact is not a small correction to the bulk ring: it is parametrically larger by $J_{zz}/(3|J_{\pm}|)$ in the controlled Ising limit.

Figure 1(a) uses the cubic-16 cell only to show the two paths in an uncluttered orthographic geometry. Every many-body result below uses FCC-32: 32 spins, 16 tetrahedra, and 2,970 ice states. Unwrapping the stored bond images gives 24 wrapping and no contractible four-cycles, followed by 32 contractible and 96 wrapping hexagons [10]. Consequently, the periodic heat-capacity maximum in Fig. 1(b) follows g_4 , not the bulk scale g_{hex} .

We now derive the projection. Write the downfolded operator as a sum of complete virtual rows, $H_{\text{eff}}^{(n)} = \sum_{r:s \rightarrow t} c_r |t\rangle \langle s|$. For a directed exchange $\ell : j \rightarrow i$, let

$$\mathbf{d}_{\ell} = \mathbf{r}_i - \mathbf{r}_j + \mathbf{n}_{\ell} \mathbf{L}, \quad V_{\ell}(\mathbf{A}) = e^{-i\mathbf{A} \cdot \mathbf{d}_{\ell}} V_{\ell}(0), \quad (4)$$

(a) cubic-16 loop geometry

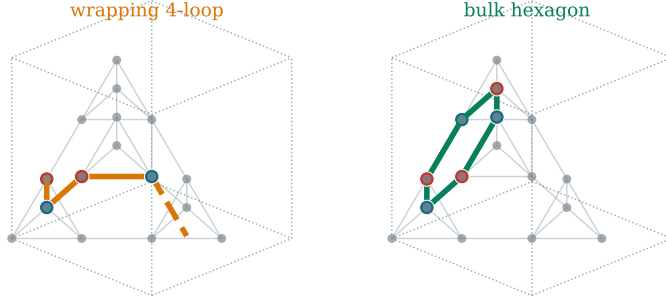
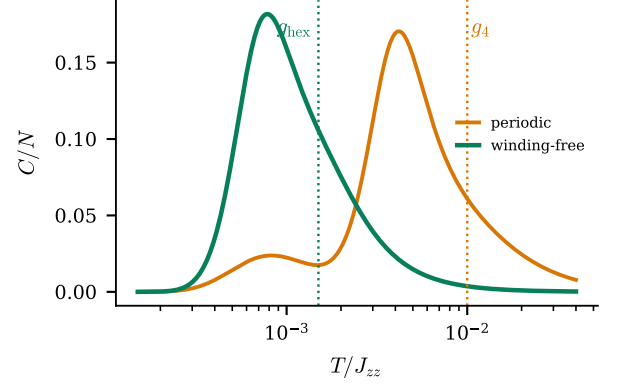
(b) scale correction, $J_{\pm}/J_{zz} = -0.05$ 

FIG. 1. **Origin and thermodynamic consequence.** (a) Orthographic views of the conventional $1 \times 1 \times 1$ cubic-16 cell. The four-site exchange closes through the boundary (dashed segment), whereas the six-site hexagon is contractible. (b) On FCC-32 at $J_{\pm}/J_{zz} = -0.05$, the periodic heat-capacity maximum follows the second-order scale g_4 ; zero-transport projection moves it to the third-order gauge scale g_{hex} .

where $\mathbf{n}_{\ell}$ records the crossed simulation-cell image. Multiplying the phases of a complete row gives

$$\prod_{\ell \in r} V_{\ell}(\mathbf{A}) = e^{-i\mathbf{A} \cdot \boldsymbol{\delta}_r} \prod_{\ell \in r} V_{\ell}(0), \quad \boldsymbol{\delta}_r = \boldsymbol{\rho}_{ts} + \mathbf{N}_r \mathbf{L}, \quad (5)$$

where $\boldsymbol{\rho}_{ts}$ is the endpoint polarization change and $\mathbf{N}_r = \sum_{\ell \in r} \mathbf{n}_{\ell}$. A change of home-cell convention shifts the two terms oppositely, making $\boldsymbol{\delta}_r$ gauge invariant. The zero translation character is therefore

$$\Pi_0 H_{\text{eff}} = \int_{\mathbb{T}_A} \frac{d^3 A}{|\mathbb{T}_A|} H_{\text{eff}}(\mathbf{A}) = \sum_{r: \boldsymbol{\delta}_r = 0} c_r |t_r\rangle \langle s_r|. \quad (6)$$

This equation also explains how bulk physics is recovered. For an alternating loop, the endpoint polarization and crossed images combine to give zero transported dipole exactly when the unwrapped loop has zero winding. Hence every contractible hexagon has $\boldsymbol{\delta}_r = 0$: all of its virtual orderings survive Eq. (6), and their sum remains g_{hex} . Every FCC-32 four-cycle has nonzero winding and is removed together with its g_4 matrix element. The projection does not manufacture a six-site process or infer a rescaled Hamiltonian; it restricts the original Schrieffer–Wolff rows to those with bulk topology.

For implementation, define the ice-state polarization $\mathbf{x}_{\alpha} = \langle \alpha | \sum_i \mathbf{r}_i S_i^z | \alpha \rangle$. Through third order on FCC-32, the minimal $\mathbb{Z}_2^3$ character average is exactly equivalent to Eq. (6) and reduces to

$$H_{\alpha\beta}^{\text{clean}} = H_{\alpha\beta} [2(\mathbf{x}_{\alpha} - \mathbf{x}_{\beta}) \text{ is componentwise even}]. \quad (7)$$

The executable protocol is: (i) enumerate the ice basis; (ii) generate every virtual row to the desired perturbative order, retaining its intermediate energy and image displacement; (iii) keep rows with $\boldsymbol{\delta}_r = 0$, or apply Eq. (7) at the order considered here; and (iv) combine retained

rows and diagonalize once. This is not an average of separately computed spectra. Ordinary boundary twists couple only to $\mathbf{N}_r$ and retain one matching of a wrapping loop, whereas Eq. (5) also contains the endpoint polarization [10]. At higher perturbative order, even nonzero winding requires a finer character grid; the row criterion Eq. (6) itself remains the definition.

Figure 2 asks directly whether the protocol also destroys physical processes. Three tests answer no. First, channel-resolved row counts show zero retention of the 24 four-loops and 96 wrapping hexagons but unit retention of all 32 contractible hexagons [Fig. 2(a)]. Second, at $J_{\pm}/J_{zz} = -0.05$, explicit $\boldsymbol{\delta} = 0$ filtering and the parity mask reduce 91,482 nonzero periodic entries to the same 18,330 entries, with zero entrywise residual in double precision. Third, varying $0.03 \leq |J_{\pm}/J_{zz}| \leq 0.10$ collapses both the full low-energy spectrum and the heat-capacity peak onto g_{hex} , giving $T_{\text{peak}}/g_{\text{hex}} = 0.52103$ [Figs. 2(b,c)]. The periodic operator cannot collapse because it contains the independent scale g_4 . Selectivity at the row level and cubic scaling of the resulting operator provide separate checks that the physical ring dynamics survives.

The same distinction is visible dynamically. To avoid introducing neutron polarization factors, we compute the longitudinal four-sublattice trace

$$S^{zz}(\mathbf{q}, \omega) = \frac{1}{N} \sum_{n,a} |\mathcal{S}_{na}(\mathbf{q})|^2 \delta[\omega - (E_n - E_0)], \quad (8)$$

$$\mathcal{S}_{na}(\mathbf{q}) = \sum_{j \in a} e^{-i\mathbf{q} \cdot \mathbf{r}_j} \langle n | S_j^z | 0 \rangle, \quad (9)$$

where a labels the four pyrochlore sublattices. This is a pure pseudospin correlation: it contains neither local-axis projections nor the laboratory neutron polarization tensor. The coherent sum $|\sum_a \mathcal{S}_{na}|^2$ vanishes at the

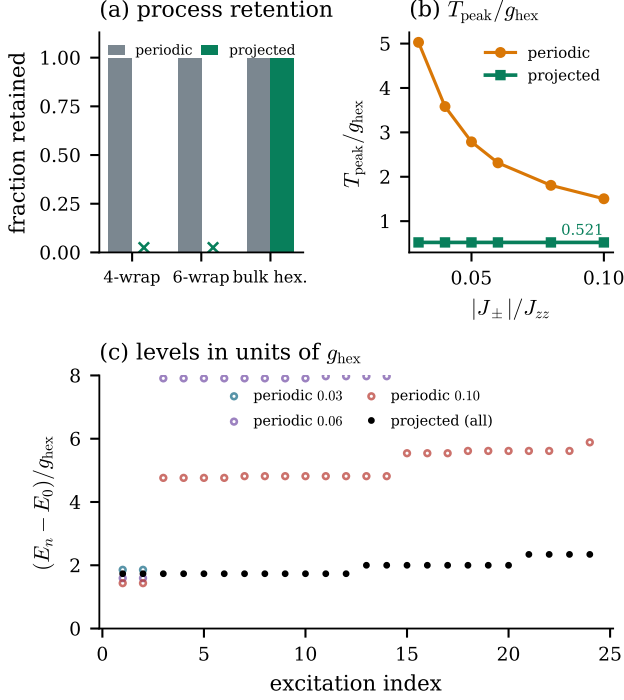


FIG. 2. **Rigorous FCC-32 benchmark.** (a) Fraction of each matrix-element channel retained. Gray bars are the periodic operator; green bars/crosses are the projected operator, which removes both wrapping channels and leaves the bulk hexagon unchanged. (b) $T_{\text{peak}}/g_{\text{hex}}$ versus $|J_{\pm}|/J_{zz}$. Projection gives the constant 0.521 expected for a Hamiltonian proportional to g_{hex} , whereas the periodic result grows because it also contains g_4 . (c) Lowest 24 excitation energies. Open colored points are periodic results at the labeled $|J_{\pm}|/J_{zz}$; black points are projected results, which coincide for every coupling when energies are divided by g_{hex} .

exact FCC-32 momenta by the ice constraint, while the sublattice trace remains finite. FCC-32 has only eight independent translation momenta: Γ , three symmetry-related X points, and four symmetry-related L points. We show only those sectors, not a continuous form-factor interpolation.

Figure 3 shows that cleaning is not a rigid energy rescaling. The artifact-selected periodic ground state has no inelastic weight in this S^{zz} trace at the allowed momenta. Projection restores finite matrix elements over the contractible-hexagon spectrum, with the first active L line at $2g_{\text{hex}}$ and cubic symmetry reproduced independently within the three X and four L sectors. The redistribution of spectral weight, together with the energy collapse in Fig. 2(c), is a dynamical check that the retained operator is not obtained by simply shifting periodic levels.

We finally apply the scale test to $\text{Ce}_2\text{Hf}_2\text{O}_7$, whose heat capacity has maxima near $T_1 = 0.065$ K and $T_2 = 0.025$ K [1]. The published seventh-order numerical-linked-cluster (NLC) fit A gives $(J_a, J_b, J_c) = (0.050, 0.021, 0.004)$ meV.

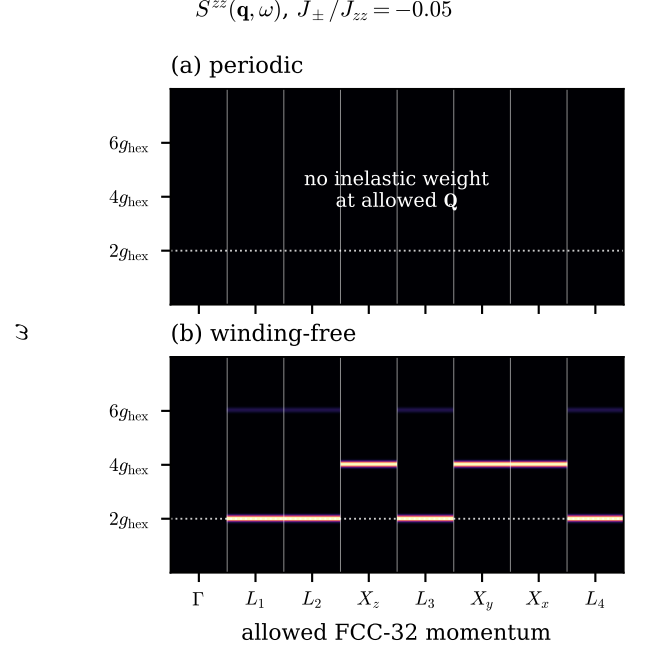


FIG. 3. **FCC-32 spectral response at the eight allowed momenta.** Periodic and winding-free $S^{zz}(\mathbf{q}, \omega)$ at $J_{\pm}/J_{zz} = -0.05$ on common linear scales. The periodic source vanishes at these momenta; the projected L response begins at the dotted $\omega = 2g_{\text{hex}}$ line and follows the contractible-ring scale.

Choosing J_a as the dominant gauge axis gives $J_{zz} = J_a$ and $J_{\pm} = -(J_b + J_c)/4 = -0.00625$ meV. The FCC-32 clean coefficient then predicts $T_{\text{ring}} = 7.1$ mK, more than a factor of three below T_2 .

The two temperatures therefore constrain different combinations of the same exchange tensor. To combine them without pretending to rerun unavailable NLC internals, we impose three explicit conditions: preserve the leading high-temperature moment $\sum_{\alpha} J_{\alpha}^2$ of set A, preserve its transverse anisotropy $J_b - J_c$, and set the clean FCC-32 ring maximum to T_2 . The unique nearby solution is

$$(J_a, J_b, J_c)_{A^*} = (0.04644, 0.02661, 0.00961) \text{ meV}, \quad (10)$$

$$J_{\pm}^* = -0.00906 \text{ meV}. \quad (11)$$

Figure 4 separates the established calculation from the proposed material inference. The published NLC-A curve describes the higher-temperature feature, but its clean low-energy ring scale misses the second peak. A^* is a better joint-scale candidate because it retains the leading high-temperature moment while satisfying the observed low-temperature constraint. A new seventh-order NLC evaluation at A^* is the decisive test of whether the full high- T line shape is also retained; no such curve is claimed here.

The central result is therefore methodological and falsifiable. On FCC-32, the leading weak-coupling error is a lower-order topological exchange, not a small shift

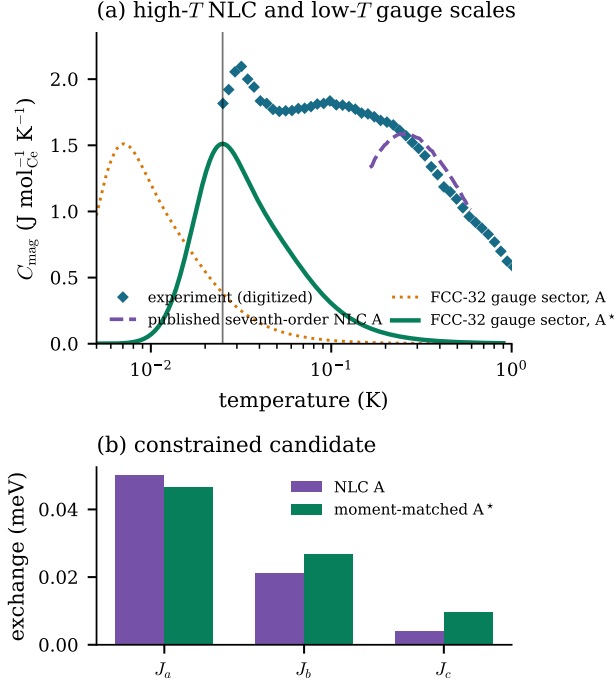


FIG. 4. **Joint constraint for $\text{Ce}_2\text{Hf}_2\text{O}_7$.** (a) Heat-capacity points and the published NLC-A curve digitized from Ref. [1] for visual comparison, with the clean FCC-32 gauge contributions for A and A*. The line marks $T_2 = 25$ mK. (b) $A^* = (0.04644, 0.02661, 0.00961)$ meV preserves $\sum_{\alpha} J_{\alpha}^2$ and $J_b - J_c$ of A while moving the clean ring peak to T_2 . The A^* bars are a parameter target, not a completed NLC curve.

of the bulk spectrum. The transported-dipole projector removes that channel exactly through third order while leaving every physical hexagon matrix element intact. Its thermodynamic and spectroscopic consequences turn the lower $\text{Ce}_2\text{Hf}_2\text{O}_7$ peak into a quantitative constraint for a joint NLC and winding-free fit.

[Acknowledgments.]

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